Solar Fe II appendix

Sun · Fe II · downloadable appendix

Document	Solar Fe II appendix
Subject	Solar photospheric abundance appendix · Fe II
Author	Ryan Schmitt
Date	2026-11-07
Publisher	The Exoplanet Codex · exoplanetcodex.org
Version	feed v1.577
Source	science commit 4cbdd50cb115
Web	https://exoplanetcodex.org/

Fe.json v1.577 · 55 Fe II products · 242 total

Solar Fe I anchor: 7.477 ± 0.049

HARPS · Reference Grade · Amarsi 3D-NLTE · n = 21. The selected Fe I product anchors this Solar appendix; products and engines are not averaged together.

Highlighted band products

near-UV

7.535 ± 0.057

Kitt Peak · Synth · 1D-LTE · Deep Grade · n = 31

Uncertainty budget closed for six of eight rows (RYA-1226); two Fe I Kitt Peak/molecfit rows keep a legacy bar pending their microturbulence bracket. On the migrated rows most of the quoted systematic is a conservative profile-width term carried from a measurement in another band.

VIS

7.398 ± 0.054

HARPS · Synth · 1D-LTE · Codex Grade · n = 145

red-optical

7.350 ± 0.048

IAG · Synth · 1D-LTE · Gerber · Reference Grade · n = 1

Line grades

Grade	What it means	This element
Reference Grade	An externally published reference line pool, using that reference’s line selection and oscillator strengths; the current Fe reference products use AGSS21. This is a provenance category, not a promise that every line has an individual primary-laboratory uncertainty.	36 products — asplund; reference
Codex Grade	The vetted Codex pool of primary-laboratory oscillator strengths, restricted to lines at or below the adopted feature-depth gate.	4 products — our-graded
Deep Grade	The corresponding laboratory-data pool above that depth gate. These stronger lines extend coverage but require particular care with saturation and blends.	15 products — our-deep-graded

Error-bar forest

How to read this forest. Each measured row is a separate abundance product for an instrument, holding, line pool and treatment; engines are never averaged together. Each bar is the uncertainty as Asplund et al. 2021 quote it: line-to-line scatter, the 3D-model term and the NLTE term (at least 0.03 dex). Atomic-data and observational terms are not in the bar, as in Asplund; they stay in the full error budget attached to the row. Green and gold regions mark the Asplund and Lodders literature values and uncertainties, not pass/fail gates. Grades describe the line pool's provenance and selection, not the quality of an engine. Reddish-orange mean-3D / Frankenstein rows are experimental and not adopted. Yellow rows are Deep Grade: strong lines that saturate, kept but less certain than the other grades. Unmeasured model rows remain N/A.

Model labels

1D-LTE uses a one-dimensional atmosphere with atomic populations in local thermodynamic equilibrium. **1D-NLTE** allows departures from that equilibrium; the named implementation supplies the correction or synthesis treatment.

Mean-3D / Frankenstein combines a spatially and temporally averaged 3D atmosphere with the Gerber synthesis route. Averaging removes the full 3D fluctuations and velocity field; this experimental hybrid is not a full 3D calculation or an adopted engine.

Synthesis fits the spectral line profile. See Methodology for the full model, line-selection and uncertainty procedures.

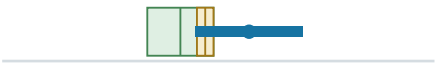
Asplund, Amarsi & Grevesse 2021 7.46 ± 0.04 · Lodders+ 2025 7.49 ± 0.01

bar = σ as Asplund 2021 quote it (scatter, 3D, NLTE) · blue = regular model result · reddish orange = experimental Frankenstein / Gerber mean-3D; not adopted · green = Asplund; gold = Lodders

near-UV — Kitt Peak — Kurucz 2005 corrected Codex deep pool
solar_kpno_kurucz2005_corrected · DEEPGRADED (no graded product in this band)
A(Fe) dex; axis 7.24–7.77

Synth · 1D-LTE Deep Grade · n=31		7.535 ±0.057
Synth · 1D-LTE · Gerber Deep Grade · n=31		7.535 ±0.058
Synth · 1D-NLTE · Bergemann Deep Grade · n=16		7.656 ±0.072

near-UV — Kitt Peak — Kurucz 2005 corrected Reference Grade · full lab pool
solar_kpno_kurucz2005_corrected
A(Fe) dex; axis 7.24–7.77

Synth · 1D-LTE Reference Grade · n=12		7.543 ±0.065
Synth · 1D-LTE · Gerber Reference Grade · n=12		7.561 ±0.066
Synth · 1D-NLTE · Bergemann Reference Grade · n=6		7.544 ±0.114
Synth · <3D>-LTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=12		7.671 ±0.066

near-UV — Kitt Peak — 1984 Molecfit corrected Codex deep pool
solar_kpno_molecfit_corrected · DEEPGRADED (no graded product in this band)
A(Fe) dex; axis 7.24–7.77

Synth · 1D-LTE Deep Grade · n=12		7.402 ±0.060
Synth · 1D-LTE · Gerber Deep Grade · n=29		7.435 ±0.063
Synth · 1D-NLTE · Bergemann Deep Grade · n=16		7.494 ±0.084

near-UV — Kitt Peak — 1984 Molecfit corrected Reference Grade · full lab pool
solar_kpno_molecfit_corrected
A(Fe) dex; axis 7.24–7.77

Synth · 1D-LTE Reference Grade · n=11		7.388 ±0.073
Synth · 1D-LTE · Gerber Reference Grade · n=12		7.386 ±0.068
Synth · 1D-NLTE · Bergemann Reference Grade · n=5		7.401 ±0.126
Synth · <3D>-LTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=12		7.489 ±0.064

near-UV axis: 7.24 · 7.38 · 7.51 · 7.64 · 7.77 A(Fe) dex

VIS — Kitt Peak — Kurucz 2005 corrected Reference Grade · Asplund replication
solar_kpno_kurucz2005_corrected
A(Fe) dex; axis 7.03–7.68

Synth · 1D-LTE Reference Grade · n=6		7.460 ±0.085
Synth · 1D-LTE · Gerber Reference Grade · n=6		7.453 ±0.085
Synth · <3D>-LTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=6		7.516 ±0.085

VIS — Kitt Peak — Kurucz 2005 corrected Codex deep pool
solar_kpno_kurucz2005_corrected · DEEPGRADED (no graded product in this band)
A(Fe) dex; axis 7.03–7.68

Synth · 1D-LTE Deep Grade · n=23		7.378 ±0.061
Synth · 1D-LTE · Gerber Deep Grade · n=23		7.386 ±0.061
Synth · 1D-NLTE · Bergemann Deep Grade · n=10		7.371 ±0.075

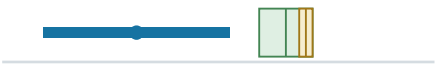
VIS — Kitt Peak — Kurucz 2005 corrected Reference Grade · full lab pool
solar_kpno_kurucz2005_corrected
A(Fe) dex; axis 7.03–7.68

Synth · 1D-LTE Reference Grade · n=8		7.454 ±0.106
Synth · 1D-LTE · Gerber Reference Grade · n=8		7.454 ±0.107
Synth · 1D-NLTE · Bergemann Reference Grade · n=2		7.232 ±0.148

VIS — Kitt Peak — 1984 Molecfit corrected Reference Grade · Asplund replication
solar_kpno_molecfit_corrected
A(Fe) dex; axis 7.03–7.68

Synth · 1D-LTE Reference Grade · n=6		7.474 ±0.080
Synth · 1D-LTE · Gerber Reference Grade · n=6		7.460 ±0.080
Synth · <3D>-LTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=6		7.522 ±0.080

VIS — Kitt Peak — 1984 Molecfit corrected Codex deep pool
solar_kpno_molecfit_corrected · DEEPGRADED (no graded product in this band)
A(Fe) dex; axis 7.03–7.68

Synth · 1D-LTE Deep Grade · n=8		7.425 ±0.087
Synth · 1D-LTE · Gerber Deep Grade · n=8		7.426 ±0.087
Synth · 1D-NLTE · Bergemann Deep Grade · n=2		7.236 ±0.140

VIS — Kitt Peak — 1984 Molecfit corrected Reference Grade · full lab pool
solar_kpno_molecfit_corrected
A(Fe) dex; axis 7.03–7.68

Synth · 1D-LTE Reference Grade · n=8		7.425 ±0.087
Synth · 1D-LTE · Gerber Reference Grade · n=8		7.426 ±0.087
Synth · 1D-NLTE · Bergemann Reference Grade · n=2		7.236 ±0.140

VIS — HARPS Reference Grade · Asplund replication solar_harps_molecfit_corrected A(Fe) dex; axis 7.03–7.68

Synth · 1D-LTE Reference Grade · n=9		7.432 ±0.064
Synth · 1D-LTE · Gerber Reference Grade · n=9		7.419 ±0.064
Synth · <3D>-LTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=9		7.474 ±0.066

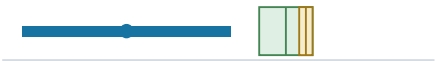
VIS — HARPS Codex deep pool solar_harps_molecfit_corrected A(Fe) dex; axis 7.03–7.68

Synth · 1D-LTE Deep Grade · n=23		7.456 ±0.081
Synth · 1D-LTE · Gerber Deep Grade · n=23		7.457 ±0.082
Synth · 1D-NLTE · Bergemann Deep Grade · n=10		7.444 ±0.083

VIS — HARPS Codex graded pool solar_harps_molecfit_corrected A(Fe) dex; axis 7.03–7.68

Synth · 1D-LTE Codex Grade · n=145		7.398 ±0.054
Synth · 1D-LTE · Gerber Codex Grade · n=145		7.390 ±0.054
Synth · 1D-NLTE · Bergemann Codex Grade · n=3		7.552 ±0.101
Synth · <3D>-LTE · Gerber · stagger · Frankenstein (experimental) Codex Grade · n=141		7.453 ±0.054

VIS — HARPS Reference Grade · full lab pool solar_harps_molecfit_corrected A(Fe) dex; axis 7.03–7.68

Synth · 1D-LTE Reference Grade · n=8		7.439 ±0.092
Synth · 1D-LTE · Gerber Reference Grade · n=8		7.443 ±0.091
Synth · 1D-NLTE · Bergemann Reference Grade · n=2		7.221 ±0.156
Synth · <3D>-LTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=8		7.531 ±0.095

VIS axis: 7.03 · 7.20 · 7.36 · 7.52 · 7.68 A(Fe) dex

red-optical — Kitt Peak — Kurucz 2005 corrected Reference Grade · Asplund replication
solar_kpno_kurucz2005_corrected
A(Fe) dex; axis 7.27–7.53

Synth · 1D-LTE Reference Grade · n=1		7.370 ±0.048
Synth · 1D-LTE · Gerber Reference Grade · n=1		7.346 ±0.048
Synth · <3D>-LTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=1		7.393 ±0.048

red-optical — Kitt Peak — 1984 Molecfit corrected Reference Grade · Asplund replication
solar_kpno_molecfit_corrected
A(Fe) dex; axis 7.27–7.53

Synth · 1D-LTE Reference Grade · n=1		7.378 ±0.049
Synth · 1D-LTE · Gerber Reference Grade · n=1		7.354 ±0.049
Synth · <3D>-LTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=1		7.401 ±0.049

red-optical — IAG Reference Grade · Asplund replication solar_iag
A(Fe) dex; axis 7.27–7.53

Synth · 1D-LTE Reference Grade · n=1		7.373 ±0.048
Synth · 1D-LTE · Gerber Reference Grade · n=1		7.350 ±0.048
Synth · <3D>-LTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=1		7.396 ±0.048

red-optical axis: 7.27 · 7.33 · 7.40 · 7.46 · 7.53 A(Fe) dex

How this was measured

Each row is one independent product. The result includes its published uncertainty (dex, Asplund 2021 convention).

Band	VIS
What was measured	Fe II · asplund
Solar spectra	HARPS solar_harps_molecfit_corrected corrected: ESO molecfit 4.4.4 per-exposure, observation-night GDAS (RYA-931)
Treatments	Synth · 1D-LTE · SYNTH · 1D-LTE · Reference Grade
Lines	9
Result A(Fe)	7.432 ± 0.064

Band	VIS
What was measured	Fe II · asplund
Uncertainty and product notes	gf scale (UNGRADED) ξ UNMEASURED INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Band	VIS
What was measured	Fe II · asplund
Solar spectra	HARPS solar_harps_molecfit_corrected corrected: ESO molecfit 4.4.4 per-exposure, observation-night GDAS (RYA-931)
Treatments	Synth · 1D-LTE · Gerber · SYNTH · synth-1D-LTE-gerber · Reference Grade
Lines	9
Result A(Fe)	7.419 ± 0.064
Uncertainty and product notes	gf scale (UNGRADED) ξ UNMEASURED INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Band	VIS
What was measured	Fe II · asplund
Solar spectra	HARPS solar_harps_molecfit_corrected corrected: ESO molecfit 4.4.4 per-exposure, observation-night GDAS (RYA-931)
Treatments	Synth · <3D>-LTE · Gerber · stagger · SYNTH · synth-mean3D-LTE-gerber-stagger · Reference Grade
Lines	9
Result A(Fe)	7.474 ± 0.066
Uncertainty and product notes	gf scale (UNGRADED) ξ UNMEASURED INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND. experimental, not adopted

Band	VIS
What was measured	Fe II · our-deep-graded
Solar spectra	HARPS solar_harps_molecfit_corrected corrected: ESO molecfit 4.4.4 per-exposure, observation-night GDAS (RYA-931)
Treatments	Synth · 1D-LTE · SYNTH · 1D-LTE · Deep Grade
Lines	23
Result A(Fe)	7.456 ± 0.081
Uncertainty and product notes	gf scale (UNGRADED) ξ MEASURED

Band	VIS
What was measured	Fe II · our-deep-graded
Solar spectra	HARPS solar_harps_molecfit_corrected corrected: ESO molecfit 4.4.4 per-exposure, observation-night GDAS (RYA-931)
Treatments	Synth · 1D-NLTE · Bergemann · SYNTH · ENGINE-A · Deep Grade
Lines	10
Result A(Fe)	7.444 ± 0.083

Band	VIS
What was measured	Fe II · our-deep-graded
Uncertainty and product notes	gf scale (UNGRADED) ξ MEASURED

Band	VIS
What was measured	Fe II · our-deep-graded
Solar spectra	HARPS solar_harps_molecfit_corrected corrected: ESO molecfit 4.4.4 per-exposure, observation-night GDAS (RYA-931)
Treatments	Synth · 1D-LTE · Gerber · SYNTH · synth-1D-LTE-gerber · Deep Grade
Lines	23
Result A(Fe)	7.457 ± 0.082
Uncertainty and product notes	gf scale (UNGRADED) ξ MEASURED

Band	VIS
What was measured	Fe II · our-graded
Solar spectra	HARPS solar_harps_molecfit_corrected corrected: ESO molecfit 4.4.4 per-exposure, observation-night GDAS (RYA-931)
Treatments	Synth · 1D-LTE · SYNTH · 1D-LTE · Codex Grade
Lines	145
Result A(Fe)	7.398 ± 0.054
Uncertainty and product notes	gf scale (UNGRADED) ξ UNMEASURED INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Band	VIS
What was measured	Fe II · our-graded
Solar spectra	HARPS solar_harps_molecfit_corrected corrected: ESO molecfit 4.4.4 per-exposure, observation-night GDAS (RYA-931)
Treatments	Synth · 1D-NLTE · Bergemann · SYNTH · ENGINE-A · Codex Grade
Lines	3
Result A(Fe)	7.552 ± 0.101
Uncertainty and product notes	gf scale (UNGRADED) ξ UNMEASURED INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Band	VIS
What was measured	Fe II · our-graded
Solar spectra	HARPS solar_harps_molecfit_corrected corrected: ESO molecfit 4.4.4 per-exposure, observation-night GDAS (RYA-931)
Treatments	Synth · 1D-LTE · Gerber · SYNTH · synth-1D-LTE-gerber · Codex Grade
Lines	145
Result A(Fe)	7.390 ± 0.054

Band	VIS
What was measured	Fe II · our-graded
Uncertainty and product notes	gf scale (UNGRADED) ξ UNMEASURED INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Band	VIS
What was measured	Fe II · our-graded
Solar spectra	HARPS solar_harps_molecfit_corrected corrected: ESO molecfit 4.4.4 per-exposure, observation-night GDAS (RYA-931)
Treatments	Synth · <3D>-LTE · Gerber · stagger · SYNTH · synth-mean3D-LTE-gerber-stagger · Codex Grade
Lines	141
Result A(Fe)	7.453 ± 0.054
Uncertainty and product notes	gf scale (UNGRADED) ξ UNMEASURED INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND. experimental, not adopted

Band	VIS
What was measured	Fe II · reference
Solar spectra	HARPS solar_harps_molecfit_corrected corrected: ESO molecfit 4.4.4 per-exposure, observation-night GDAS (RYA-931)
Treatments	Synth · 1D-LTE · SYNTH · 1D-LTE · Reference Grade
Lines	8
Result A(Fe)	7.439 ± 0.092
Uncertainty and product notes	line-to-line scatter ξ MEASURED

Band	VIS
What was measured	Fe II · reference
Solar spectra	HARPS solar_harps_molecfit_corrected corrected: ESO molecfit 4.4.4 per-exposure, observation-night GDAS (RYA-931)
Treatments	Synth · 1D-NLTE · Bergemann · SYNTH · ENGINE-A · Reference Grade
Lines	2
Result A(Fe)	7.221 ± 0.156
Uncertainty and product notes	line-to-line scatter ξ UNMEASURED INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Band	VIS
What was measured	Fe II · reference
Solar spectra	HARPS solar_harps_molecfit_corrected corrected: ESO molecfit 4.4.4 per-exposure, observation-night GDAS (RYA-931)
Treatments	Synth · 1D-LTE · Gerber · SYNTH · synth-1D-LTE-gerber · Reference Grade
Lines	8
Result A(Fe)	7.443 ± 0.091

Band	VIS
What was measured	Fe II · reference
Uncertainty and product notes	line-to-line scatter ξ MEASURED

Band	VIS
What was measured	Fe II · reference
Solar spectra	HARPS solar_harps_molecfit_corrected corrected: ESO molecfit 4.4.4 per-exposure, observation-night GDAS (RYA-931)
Treatments	Synth · <3D>-LTE · Gerber · stagger · SYNTH · synth-mean3D-LTE-gerber-stagger · Reference Grade
Lines	8
Result A(Fe)	7.531 ± 0.095
Uncertainty and product notes	line-to-line scatter ξ UNMEASURED INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND. experimental, not adopted

Band	VIS
What was measured	Fe II · asplund
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · 1D-LTE · SYNTH · 1D-LTE · Reference Grade
Lines	6
Result A(Fe)	7.460 ± 0.085
Uncertainty and product notes	gf scale (UNGRADED) ξ MEASURED

Band	VIS
What was measured	Fe II · asplund
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · 1D-LTE · Gerber · SYNTH · synth-1D-LTE-gerber · Reference Grade
Lines	6
Result A(Fe)	7.453 ± 0.085
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Band	VIS
What was measured	Fe II · asplund
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · <3D>-LTE · Gerber · stagger · SYNTH · synth-mean3D-LTE-gerber-stagger · Reference Grade
Lines	6
Result A(Fe)	7.516 ± 0.085
Uncertainty and product notes	gf scale (UNGRADED) ξ MEASURED experimental, not adopted

Band	VIS
What was measured	Fe II · our-deep-graded
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · 1D-LTE · SYNTH · 1D-LTE · Deep Grade
Lines	23
Result A(Fe)	7.378 ± 0.061
Uncertainty and product notes	gf scale (UNGRADED) ξ MEASURED

Band	VIS
What was measured	Fe II · our-deep-graded
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · 1D-NLTE · Bergemann · SYNTH · ENGINE-A · Deep Grade
Lines	10
Result A(Fe)	7.371 ± 0.075
Uncertainty and product notes	gf scale (UNGRADED) ξ MEASURED

Band	VIS
What was measured	Fe II · our-deep-graded
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · 1D-LTE · Gerber · SYNTH · synth-1D-LTE-gerber · Deep Grade
Lines	23
Result A(Fe)	7.386 ± 0.061
Uncertainty and product notes	gf scale (UNGRADED) ξ MEASURED

Band	VIS
What was measured	Fe II · reference
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · 1D-LTE · SYNTH · 1D-LTE · Reference Grade
Lines	8
Result A(Fe)	7.454 ± 0.106
Uncertainty and product notes	line-to-line scatter ξ MEASURED

Band	VIS
What was measured	Fe II · reference
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · 1D-NLTE · Bergemann · SYNTH · ENGINE-A · Reference Grade
Lines	2

Band	VIS
What was measured	Fe II · reference
Result A(Fe)	7.232 ± 0.148
Uncertainty and product notes	line-to-line scatter ξ UNMEASURED INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Band	VIS
What was measured	Fe II · reference
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · 1D-LTE · Gerber · SYNTH · synth-1D-LTE-gerber · Reference Grade
Lines	8
Result A(Fe)	7.454 ± 0.107
Uncertainty and product notes	line-to-line scatter ξ MEASURED

Band	VIS
What was measured	Fe II · asplund
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · 1D-LTE · SYNTH · 1D-LTE · Reference Grade
Lines	6
Result A(Fe)	7.474 ± 0.080
Uncertainty and product notes	gf scale (UNGRADED) ξ MEASURED

Band	VIS
What was measured	Fe II · asplund
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · 1D-LTE · Gerber · SYNTH · synth-1D-LTE-gerber · Reference Grade
Lines	6
Result A(Fe)	7.460 ± 0.080
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Band	VIS
What was measured	Fe II · asplund
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · <3D>-LTE · Gerber · stagger · SYNTH · synth-mean3D-LTE-gerber-stagger · Reference Grade
Lines	6
Result A(Fe)	7.522 ± 0.080
Uncertainty and product notes	gf scale (UNGRADED) ξ MEASURED experimental, not adopted

Band	VIS
What was measured	Fe II · our-deep-graded
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · 1D-LTE · SYNTH · 1D-LTE · Deep Grade
Lines	8
Result A(Fe)	7.425 ± 0.087
Uncertainty and product notes	line-to-line scatter ξ MEASURED

Band	VIS
What was measured	Fe II · our-deep-graded
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · 1D-NLTE · Bergemann · SYNTH · ENGINE-A · Deep Grade
Lines	2
Result A(Fe)	7.236 ± 0.140
Uncertainty and product notes	line-to-line scatter ξ UNMEASURED INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Band	VIS
What was measured	Fe II · our-deep-graded
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · 1D-LTE · Gerber · SYNTH · synth-1D-LTE-gerber · Deep Grade
Lines	8
Result A(Fe)	7.426 ± 0.087
Uncertainty and product notes	line-to-line scatter ξ MEASURED

Band	VIS
What was measured	Fe II · reference
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · 1D-LTE · SYNTH · 1D-LTE · Reference Grade
Lines	8
Result A(Fe)	7.425 ± 0.087
Uncertainty and product notes	line-to-line scatter ξ MEASURED

Band	VIS
What was measured	Fe II · reference
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)

Band	VIS
What was measured	Fe II · reference
Treatments	Synth · 1D-NLTE · Bergemann · SYNTH · ENGINE-A · Reference Grade
Lines	2
Result A(Fe)	7.236 ± 0.140
Uncertainty and product notes	line-to-line scatter ξ UNMEASURED INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Band	VIS
What was measured	Fe II · reference
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · 1D-LTE · Gerber · SYNTH · synth-1D-LTE-gerber · Reference Grade
Lines	8
Result A(Fe)	7.426 ± 0.087
Uncertainty and product notes	line-to-line scatter ξ MEASURED

Band	near-UV
What was measured	Fe II · our-deep-graded
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · 1D-LTE · SYNTH · 1D-LTE · Deep Grade
Lines	31
Result A(Fe)	7.535 ± 0.057
Uncertainty and product notes	gf scale (UNGRADED) ξ MEASURED

Band	near-UV
What was measured	Fe II · our-deep-graded
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · 1D-NLTE · Bergemann · SYNTH · ENGINE-A · Deep Grade
Lines	16
Result A(Fe)	7.656 ± 0.072
Uncertainty and product notes	gf scale (UNGRADED) ξ MEASURED

Band	near-UV
What was measured	Fe II · our-deep-graded
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · 1D-LTE · Gerber · SYNTH · synth-1D-LTE-gerber · Deep Grade
Lines	31
Result A(Fe)	7.535 ± 0.058

Band	near-UV
What was measured	Fe II · our-deep-graded
Uncertainty and product notes	gf scale (UNGRADED) ξ MEASURED

Band	near-UV
What was measured	Fe II · reference
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · 1D-LTE · SYNTH · 1D-LTE · Reference Grade
Lines	12
Result A(Fe)	7.543 ± 0.065
Uncertainty and product notes	pseudo-continuum ξ MEASURED

Band	near-UV
What was measured	Fe II · reference
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · 1D-NLTE · Bergemann · SYNTH · ENGINE-A · Reference Grade
Lines	6
Result A(Fe)	7.544 ± 0.114
Uncertainty and product notes	line-to-line scatter ξ MEASURED

Band	near-UV
What was measured	Fe II · reference
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · 1D-LTE · Gerber · SYNTH · synth-1D-LTE-gerber · Reference Grade
Lines	12
Result A(Fe)	7.561 ± 0.066
Uncertainty and product notes	pseudo-continuum ξ MEASURED

Band	near-UV
What was measured	Fe II · reference
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · <3D>-LTE · Gerber · stagger · SYNTH · synth-mean3D-LTE-gerber-stagger · Reference Grade
Lines	12
Result A(Fe)	7.671 ± 0.066
Uncertainty and product notes	pseudo-continuum ξ MEASURED experimental, not adopted

Band	near-UV
What was measured	Fe II · our-deep-graded
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · 1D-LTE · SYNTH · 1D-LTE · Deep Grade
Lines	12
Result A(Fe)	7.402 ± 0.060
Uncertainty and product notes	pseudo-continuum ξ MEASURED

Band	near-UV
What was measured	Fe II · our-deep-graded
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · 1D-NLTE · Bergemann · SYNTH · ENGINE-A · Deep Grade
Lines	16
Result A(Fe)	7.494 ± 0.084
Uncertainty and product notes	gf scale (UNGRADED) ξ MEASURED

Band	near-UV
What was measured	Fe II · our-deep-graded
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · 1D-LTE · Gerber · SYNTH · synth-1D-LTE-gerber · Deep Grade
Lines	29
Result A(Fe)	7.435 ± 0.063
Uncertainty and product notes	gf scale (UNGRADED) ξ MEASURED

Band	near-UV
What was measured	Fe II · reference
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · 1D-LTE · SYNTH · 1D-LTE · Reference Grade
Lines	11
Result A(Fe)	7.388 ± 0.073
Uncertainty and product notes	pseudo-continuum ξ MEASURED

Band	near-UV
What was measured	Fe II · reference
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · 1D-NLTE · Bergemann · SYNTH · ENGINE-A · Reference Grade
Lines	5

Band	near-UV
What was measured	Fe II · reference
Result A(Fe)	7.401 ± 0.126
Uncertainty and product notes	line-to-line scatter ξ MEASURED

Band	near-UV
What was measured	Fe II · reference
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · 1D-LTE · Gerber · SYNTH · synth-1D-LTE-gerber · Reference Grade
Lines	12
Result A(Fe)	7.386 ± 0.068
Uncertainty and product notes	pseudo-continuum ξ MEASURED

Band	near-UV
What was measured	Fe II · reference
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · <3D>-LTE · Gerber · stagger · SYNTH · synth-mean3D-LTE-gerber-stagger · Reference Grade
Lines	12
Result A(Fe)	7.489 ± 0.064
Uncertainty and product notes	pseudo-continuum ξ MEASURED experimental, not adopted

Band	red-optical
What was measured	Fe II · asplund
Solar spectra	IAG solar_iag corrected: as delivered — measured telluric-free to 0.1% in the O2 A-band core (RYA-783/786)
Treatments	Synth · 1D-LTE · SYNTH · 1D-LTE · Reference Grade
Lines	1
Result A(Fe)	7.373 ± 0.048
Uncertainty and product notes	gf scale (UNGRADED) ξ UNMEASURED INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Band	red-optical
What was measured	Fe II · asplund
Solar spectra	IAG solar_iag corrected: as delivered — measured telluric-free to 0.1% in the O2 A-band core (RYA-783/786)
Treatments	Synth · 1D-LTE · Gerber · SYNTH · synth-1D-LTE-gerber · Reference Grade
Lines	1
Result A(Fe)	7.350 ± 0.048

Band	red-optical
What was measured	Fe II · asplund
Uncertainty and product notes	gf scale (UNGRADED) ξ UNMEASURED INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Band	red-optical
What was measured	Fe II · asplund
Solar spectra	IAG solar_iag corrected: as delivered — measured telluric-free to 0.1% in the O2 A-band core (RYA-783/786)
Treatments	Synth · <3D>-LTE · Gerber · stagger · SYNTH · synth-mean3D-LTE-gerber-stagger · Reference Grade
Lines	1
Result A(Fe)	7.396 ± 0.048
Uncertainty and product notes	gf scale (UNGRADED) ξ UNMEASURED INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND. experimental, not adopted

Band	red-optical
What was measured	Fe II · asplund
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · 1D-LTE · SYNTH · 1D-LTE · Reference Grade
Lines	1
Result A(Fe)	7.370 ± 0.048
Uncertainty and product notes	gf scale (UNGRADED) ξ UNMEASURED INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Band	red-optical
What was measured	Fe II · asplund
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · 1D-LTE · Gerber · SYNTH · synth-1D-LTE-gerber · Reference Grade
Lines	1
Result A(Fe)	7.346 ± 0.048
Uncertainty and product notes	gf scale (UNGRADED) ξ UNMEASURED INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Band	red-optical
What was measured	Fe II · asplund
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · <3D>-LTE · Gerber · stagger · SYNTH · synth-mean3D-LTE-gerber-stagger · Reference Grade
Lines	1

Band	red-optical
What was measured	Fe II · asplund
Result A(Fe)	7.393 ± 0.048
Uncertainty and product notes	gf scale (UNGRADED) ξ UNMEASURED INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND. experimental, not adopted

Band	red-optical
What was measured	Fe II · asplund
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · 1D-LTE · SYNTH · 1D-LTE · Reference Grade
Lines	1
Result A(Fe)	7.378 ± 0.049
Uncertainty and product notes	gf scale (UNGRADED) ξ UNMEASURED INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Band	red-optical
What was measured	Fe II · asplund
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · 1D-LTE · Gerber · SYNTH · synth-1D-LTE-gerber · Reference Grade
Lines	1
Result A(Fe)	7.354 ± 0.049
Uncertainty and product notes	gf scale (UNGRADED) ξ UNMEASURED INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Band	red-optical
What was measured	Fe II · asplund
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · <3D>-LTE · Gerber · stagger · SYNTH · synth-mean3D-LTE-gerber-stagger · Reference Grade
Lines	1
Result A(Fe)	7.401 ± 0.049
Uncertainty and product notes	gf scale (UNGRADED) ξ UNMEASURED INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND. experimental, not adopted

Solar iron context

Fe II traces singly ionized iron and is retained as an ionization and consistency diagnostic alongside Fe I, not an independent headline to average with it. The current resolved pool contains strong, saturation-sensitive lines and documented blend exclusions; the diagnostics below distinguish those limitations from model and holding differences.

Near-UV opacity — report and noted

Molecular opacity INCLUDED — OH A-X, NH, CH and CN violet, 6,223 VALD lines as Turbospectrum lists with the dominant isotopologue (RYA-1207). The band previously carried none: every molecular list iSpec ships begins at 400 nm. Measured PAIRED against a molecules-off run on the same code and pool, the lever is -0.047..-0.055 dex on Fe I and -0.276..-0.332 on Fe II, with chi2r median 81.8 -> 67.4 and control windows containing no molecular line bit-identical. RYA-1209 additionally curated out Fe I 3721.272 (systematic:K07 — gf attributed to a VALD3 reference that publishes no per-line uncertainty, no NIST accuracy class, so it never met the DEEPGRADED tier's lab-gf definition); the Fe I pool is 57 candidates and the budget clears at gf rung 3 (cited lab), 55/55 GF-LAB, RMS 0.0509 dex. RESIDUAL DEFICIT: the correction still UNDER-corrects — RYA-1204 measured synth/obs above 1 on 24 of 40 windows after it — so the band remains short of opacity by an amount not yet measured on this pool, and these values are a floor rather than a converged answer. sigma_syst is unchanged at 0.1122 because it is dominated by the 0.10 dex pseudo-continuum term, not by gf: the true continuum is never observed here. Not adopted for the headline. The headline is the per-band best-available product, selected from this feed by the publication generator rather than fixed as a constant: RYA-850 promoted the graded laboratory-gf pool to the PRIMARY reported value, and RYA-819 retired the earlier constructed anchor after showing its Magic-2013 correction wrong in both magnitude and shape. RYA-161: nothing calibrated to a target.

Frankenstein — experimental, pending the Bride

48 experimental products remain available in the Fe I product values and downloads.

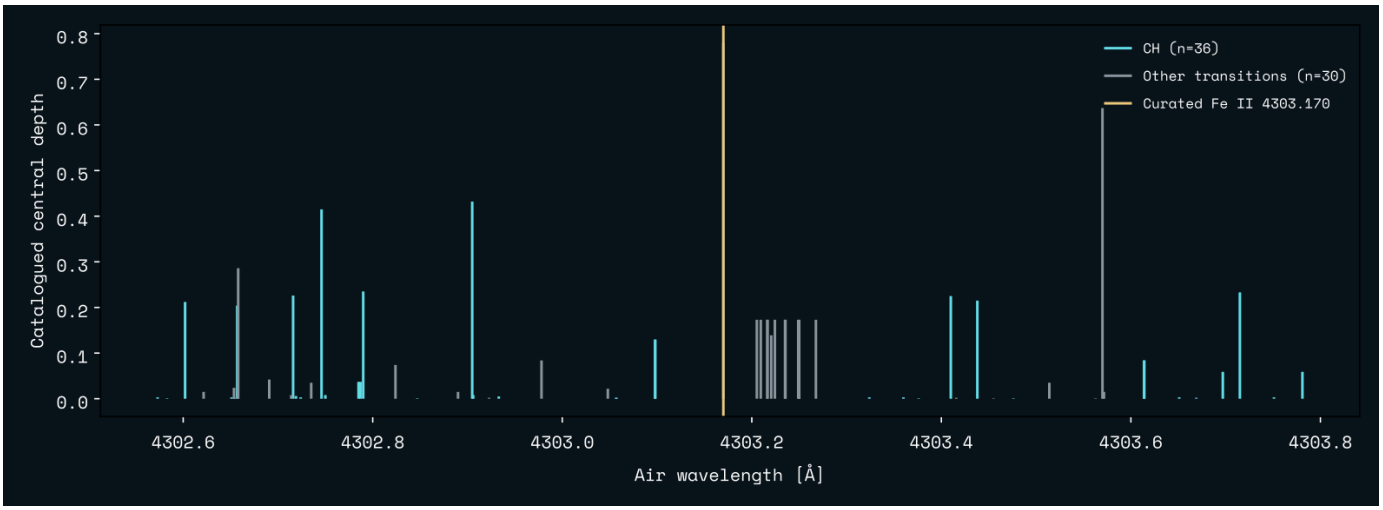
RYA-1197 Part A: the Frankenstein mean-<3D> route is HELD FROM HEADLINE publication until the Bride validates it. It inflates A(Fe) by roughly +0.10 dex against the 1D siblings and exists on VIS/KP only, so it is not a comparable measurement of the same quantity. It stays in the feed deliberately: the LTE/NLTE mean-3D PAIR is the NLTE differential on ONE atmosphere (RYA-542), which is the diagnostic it is for. Not a headline value.

Saturation characterization

Committed proof census: 120 line/holding records; 100% have observed depth > 0.70. Median observed log10(EW[mÅ]/λ[Å]) = -1.252. In consistent Å units this is -4.252. These diagnostics characterize the audited saturated pool, not every newly added product.

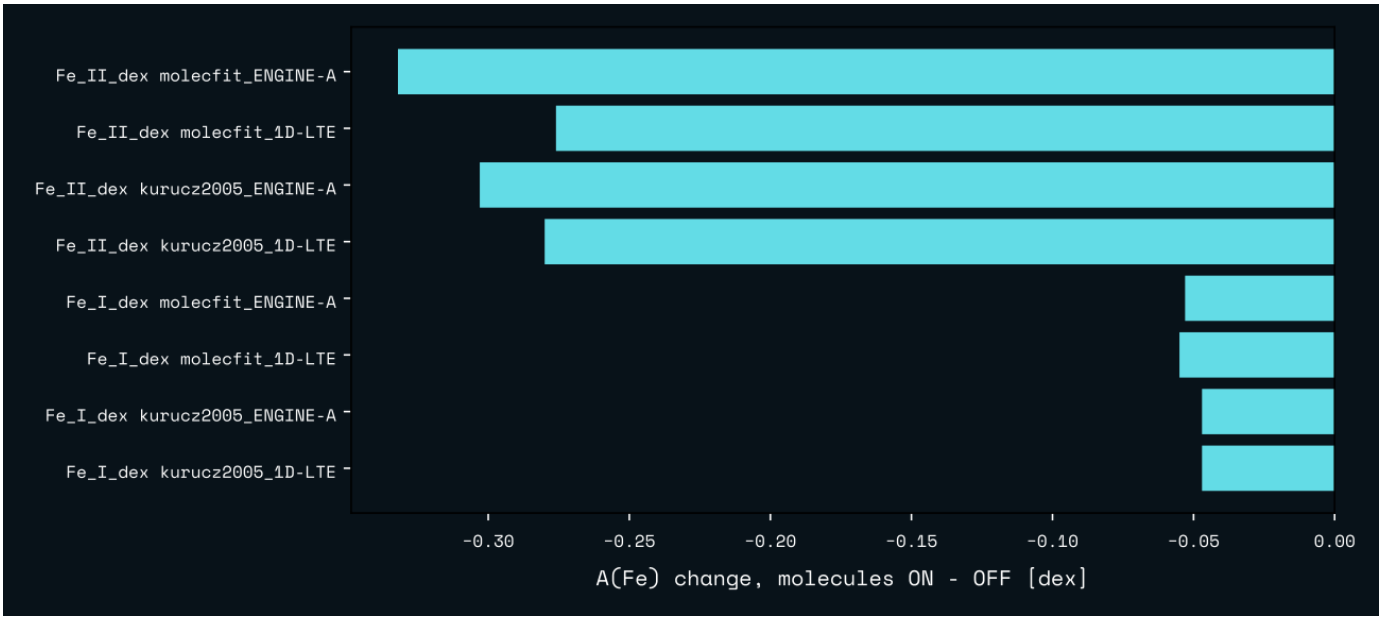
Problem-line and model diagnostics

CH G-band artifact



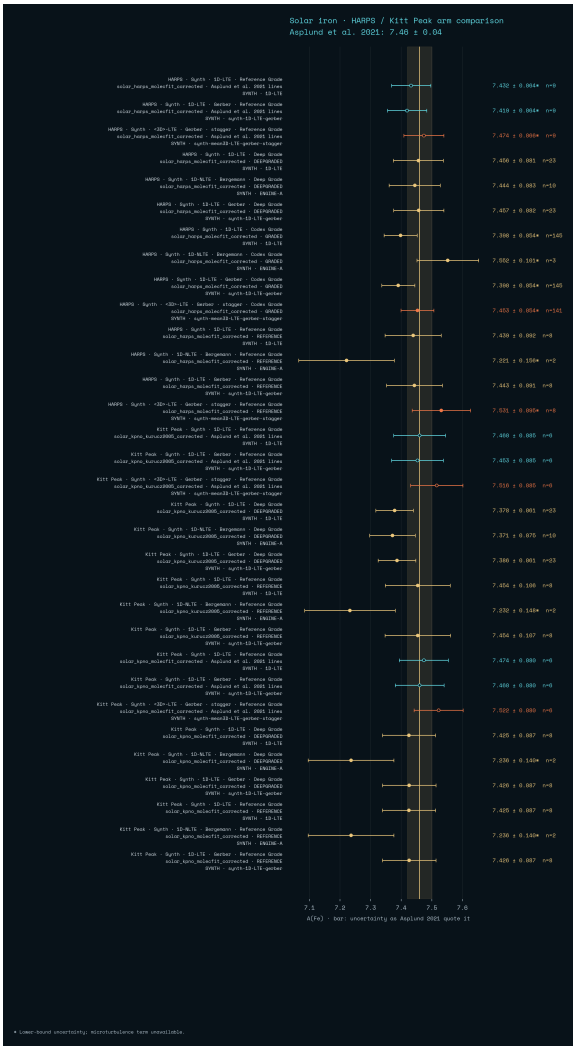
CH G-band blend artifact — the synthesis list cannot represent the dominant absorber · A = 10.0 (+2.5 dex). RYA-1191 blend audit: 66 catalogued transitions in the 0.62 A fit window, top contaminant CH I at 43.8% of the summed depth against Fe II's 11.1%. The VIS synthesis list GESv6_atom_hfs_iso carries 206 species and NOT ONE molecular, so the fit had nowhere to put the CH absorption except Fe II. RYA-515 independently counted 36 of 65 transitions as CH. The plot shows the committed transition census in the proof generator's fit window; it is not a synthetic spectrum. Source evidence

Near-UV molecular opacity



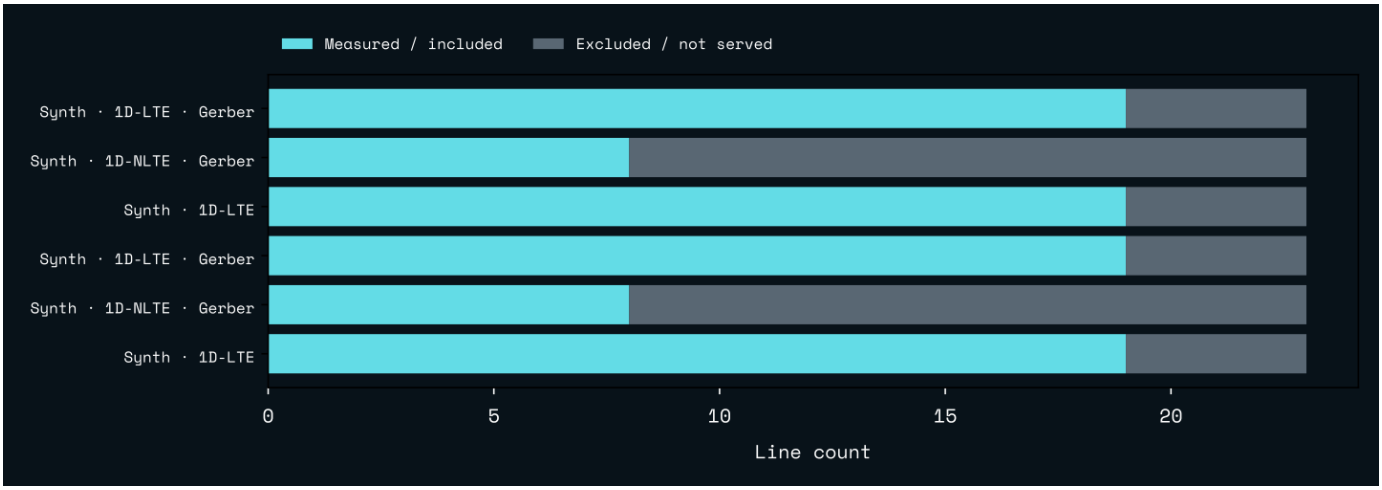
molecules OFF vs ON on THE SAME CODE and the same pool. Differencing against the published values instead would have measured the molecular lever PLUS every code change since the feed was written on Sept 3 -- which is about -0.035 dex of the total. Source evidence

HARPS arm offset



Current Fe II VIS products, separated by holding and engine. The arm offset persists after the curated CH artifact was removed. Source evidence

CRIRES+ H model coverage



Included and excluded line counts from each current product. Excluded rows are not automatically failed fits; the per-line reason distinguishes unavailable NLTE coverage. Source evidence

Download the evidence

1341 graded per-line measurements exported. Evidence for 152 graded products is missing or does not reproduce the feed; 0 exported lines lack an unambiguous gf match. The download is explicitly incomplete until those upstream records are supplied.

Graded per-line abundances (CSV)

All product fields (CSV; nested fields preserved as JSON)

Per-product line-evidence coverage (CSV)

Source Fe.json

Reproducibility

Generator scripts/generate_fe_publication.py v1.0.0
Science source commit 4cbdd50cb115757aeacaa83122f1ae28489868b8
Fe.json v1.577 · feed timestamp 2026-11-06T23:24:27Z
Generated 2026-11-07T00:03:25+00:00
Feed SHA-256 6bcdb992975a7c62351c0ed00dfbf850ef1d52e217f9182c17c37a6d180143b7

Refreshed product tracker · Build manifest · Fe I appendix

References

Asplund, M.; Amarsi, A. M.; Grevesse, N. (2021). The chemical make-up of the Sun: A 2020 vision (doi: 10.1051/0004-6361/202140445). AGSS21 Solar reference and the green literature comparison; Table A.2 defines the Reference Grade Fe I line set.

Lodders, K.; Bergemann, M.; Palme, H. (2025). Solar System Elemental Abundances from the Solar Photosphere and CI-Chondrites (doi: 10.1007/s11214-025-01146-w). Table 6 present-day Solar composition, shown as the gold literature comparison.

Gerber, J. M.; Magg, E.; Plez, B.; Bergemann, M.; Heiter, U.; Olander, T.; Hoppe, R. (2023). Non-LTE radiative transfer with Turbospectrum (doi: 10.1051/0004-6361/202243673). The Gerber Turbospectrum non-LTE synthesis implementation named in the model rows.

Den Hartog, E. A.; Lawler, J. E.; Sneden, C.; Cowan, J. J.; Brukhovesky, A. (2019). Atomic Transition Probabilities for UV and Blue Lines of Fe II and Abundance Determinations in the Photospheres of the Sun and Metal-poor Star HD 84937 (doi: 10.3847/1538-4365/ab322e). Laboratory transition probabilities used in the resolved Fe II per-line evidence. Coverage and unresolved products are listed in the downloads.